

PCA

PCA (Principal Component Analysis)

- ↳ It is dimensionality reduction technique
- ↳ Reduce the no. of features in dataset.
- ↳ It change retain the most important features.
- ↳ It changes the complex datasets by transforming correlated features into a small set of uncorrelated components.
- ↳ PCA uses linear algebra to transform data into new feature called principal components.
- ↳ It finds these by calculating eigenvector (directions) and eigenvalue (importance) from the covariance matrix.
- ↳ PCA takes selects the top components with highest eigenvalues and project data them simplify the dataset.
- ↳ It is also known as curse of dimensionality.
- ↳

1. Standardize the data

↳ Different features may have different units and scales like salary vs age.

↳ To compare them fairly PCA first standardize the data by making each feature have

1. mean of 0

2. standard deviation of 1

$$Z = \frac{x - \mu}{\sigma}$$

where μ is the mean of independent

$$\mu = (\mu_1, \mu_2, \dots, \mu_m)$$

σ is the standard deviation of independent feature $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_m)$

2. Calculate covariance matrix

↳ PCA calculate the covariance matrix to see how feature relate to each other whether they increase or decrease together.

↳ Covariance b/w two features x_1 & x_2

$$\text{Cov}(x_1, x_2) = \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1}$$

• where $\bar{x}_1$ & $\bar{x}_2$ are mean values of features x_1 & x_2

• n is the number of data points.

↳ The value of covariance can be positive, negative or zero

3. Find the Principal Component

↳ PCA identifies new axes where the data spreads out most

↳ 1st Principal Component (PC1): The direction of main maximum variance (most spread)

↳ 2nd Principal Component (PC2) The best direction perpendicular to PC1

↳ These direction come from the eigenvectors of covariance matrix and their importance is measured by eigenvalues.

↳ For a square matrix A an eigenvector x and its corresponding eigenvalue λ satisfy

$$Ax = \lambda x$$

This means:

- when A acts on x it only stretches or shrinks x by scalar λ

- The direction of x remains unchanged hence eigenvector defines "stable direction" of A .

↳ Eigenvalues help rank these direction by importance

4 * Pick the top direction and Transform Data.

↳ After calculating the eigenvectors and eigenvalues PCA rank them by the amount of information they capture we then:

1. Select the top K components that capture most of the variance like 95%.
2. Transform the original dataset by projecting it onto these top components.

This means we reduce the no. of features while keeping the important patterns in the data.

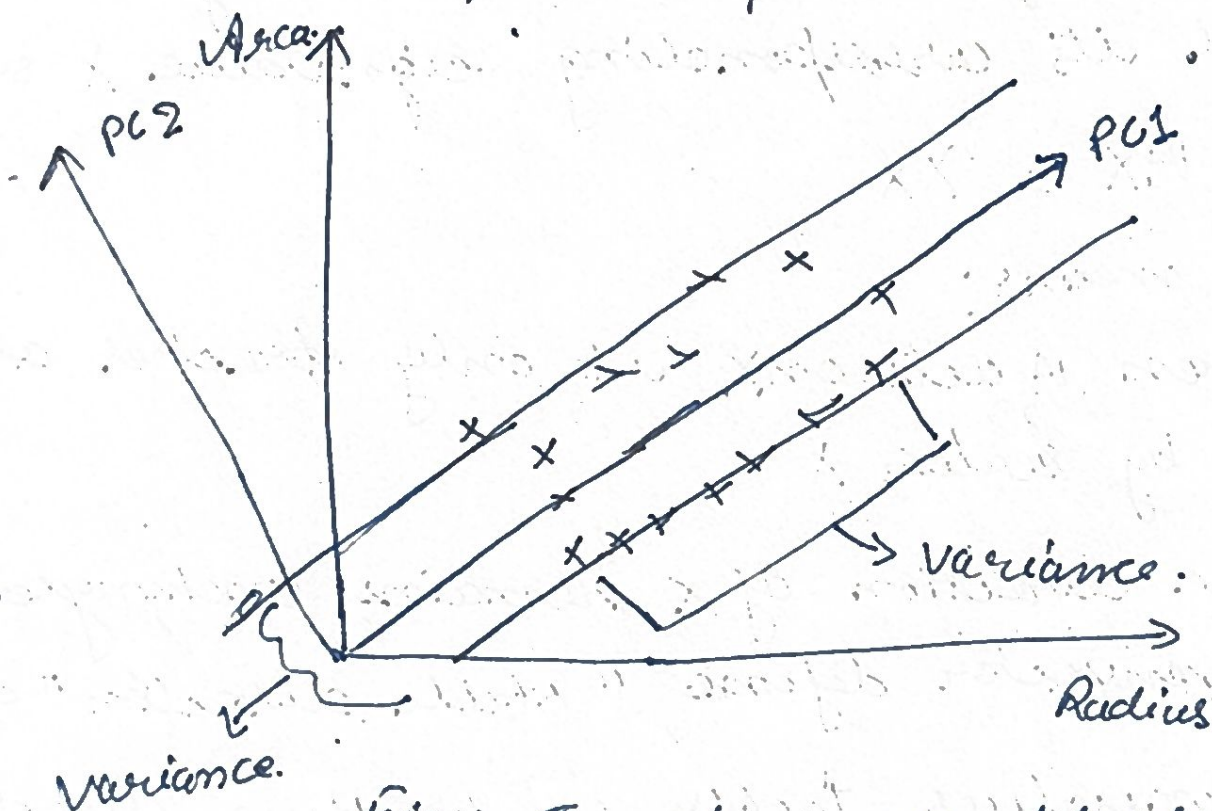


Fig: Transform 2D dataset into 1D

- PC1: capture ^{maximum} variance i.e. it hold most information while PC2 capture remaining variance and perpendicular to PC1
- The spread of data is much wide along PC1 than along PC2. Therefore PC1 chosen for dimensionality reduction.
- Projecting the data points onto PC1, we effectively transform the 2D data into 1D

Step for PCA

1) Data preparation

↳ collect and preprocess i.e. standardized data.

2) Covariance Matrix computation.

3) Compute eigenvalue and eigenvector.

↳ Calculate the eigenvalues and eigenvectors of covariance matrix. These represent the magnitude and direction of principal components. These eigenvalues indicate the variance explained by each principal component, and the eigenvectors define the dirⁿ of these component.

4. Sort eigenvalues:

↳ Sort the eigenvalues in descending order.

↳ largest eigenvalues correspond to the most significant principal components.

5. Select Principal components -

↳ Decide how many principal components to keep,

↳ we can select a specific no. of or ~~choose~~ choose based on explained var (keep components that explain 95% of the variance).

6. Projection onto Principal components -

↳ Project your original data onto the selected principal components to obtain a new set of features that are linear ~~compo~~ combination of the original features.

Convert 2D to 1D data using PCA
 Step 1 - Collect the data

Feature	E_1	E_2	E_3	E_4
x_1	2	3	4	5
x_2	4	6	8	10

$$\bar{x}_1 = \frac{2+3+4+5}{4} = \frac{14}{4} = 3.5$$

$$\bar{x}_2 = \frac{4+6+8+10}{4} = \frac{28}{4} = 7$$

Step 2: Covariance.

$$S = \frac{1}{N} \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) \end{bmatrix}$$

$$\text{Cov}(x_1, x_1) = \frac{1}{N-1} \sum_{i=1}^N (x_{1i} - \bar{x}_1)^2$$

$$= \frac{1}{3} \left[(2-3.5)^2 + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2 \right]$$

$$= \frac{1}{3} = 1.83$$

$$\text{Cov}(x_1, x_2) = \frac{1}{N-1} \sum_{i=1}^N (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)$$

$$= \frac{1}{3} \left[(2-3.5)(4-7) + (3-3.5)(6-7) + (4-3.5)(8-7) + (5-3.5)(10-7) \right]$$

$$= 3.33$$

$$\text{Cov}(x_1, x_2) = \text{Cov}(x_2, x_1) = 3.33$$

$$\text{Cov}(x_2 - \bar{x}_2) = \frac{1}{N-1} \sum_{i=1}^N (x_{2i} - \bar{x}_2)^2$$

$$= \frac{1}{3} (4-7)^2 + (6-7)^2 + (8-7)^2 + (10-7)^2$$

$$= \frac{1}{3} [9 + 1 + 1 + 9]$$

$$= \frac{20}{3} = 6.67$$

$$\text{Cov matrix} = \begin{bmatrix} 1.83 & 3.33 \\ 3.33 & 6.67 \end{bmatrix}$$

Step 3: Eigen values and eigenvector

$$|S - \lambda I| = 0$$

$$\begin{vmatrix} 1.83 - \lambda & 3.33 \\ 3.33 & 6.67 - \lambda \end{vmatrix} = 0$$

$$\lambda^2 - 8.5\lambda + 1.12 = 0$$

Solve above equation we get.

$$\lambda = 8.37, 0.14$$

Now calculate eigenvector

$$V = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$(S - \lambda I)V = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1.83 - \lambda_1 & 3.33 \\ 3.33 & 6.67 - \lambda_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} (1.83 - \lambda_1)u_1 + 3.33u_2 \\ 3.33u_1 + (6.67 - \lambda_1)u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(1.83 - \lambda_1)u_1 + 3.33u_2 = 0$$

$$3.33u_1 + (6.67 - \lambda_1)u_2 = 0$$

$$\frac{u_1}{-3.33} = \frac{u_2}{1.83 - \lambda_1} = t$$

$$u_1 = -3.33t, \quad u_2 = (1.83 - \lambda_1)t$$

Consider $t = 1$

$$V = \begin{bmatrix} -3.33 \\ 1.83 - \lambda_1 \end{bmatrix}$$

Step 4:

Sort eigen vector values.

↳ To calculate unit eigen vector, we have.

to calculate the length of U_1 .

↳ To obtain unit eigenvector, we normalized a vector, divided each component by its magnitude.

$$\begin{aligned} \|U_1\| &= \sqrt{(-3.33)^2 + (1.83 - 8.37)^2} \\ &= 7.34 \end{aligned}$$

$$e_1 = \begin{bmatrix} \frac{(-3.33)}{7.34} \\ \frac{(1.83 - 8.37)}{7.34} \end{bmatrix} = \begin{bmatrix} -0.45 \\ -0.89 \end{bmatrix}$$

Now Compute for λ_2

Similarly compute as e_1 we get

$$e_2 = \begin{bmatrix} -0.89 \\ 0.45 \end{bmatrix}$$

Step 5. Form feature vector:

$$\begin{bmatrix} -0.45 & -0.89 \\ -0.89 & 0.45 \end{bmatrix}$$

Step 5 select first principal component.

equation to calculate the principal component is as follows.

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix}$$

apply first eigenvector, we will get the following.

$$\begin{bmatrix} -0.45 & -0.89 \end{bmatrix} \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix}$$

$$= -0.45 (x_{1k} - \bar{x}_1) - 0.89 (x_{2k} - \bar{x}_2)$$

Step 6 Projection onto principal components.

for point (2, 4)

$$= -0.45 (2 - 3.5) - 0.89 (4 - 7)$$

$$= 0.68 + 2.67$$

for point (3,6)

$$= -0.45(3 - 3.5) - 0.89(6 - 7)$$

$$= 0.23 + 0.89$$

$$= 1.12$$

similar for (4,8) we get -1.12 & (5,10)

we get $= -3.35$

thus final answer.

x_1	2	3	4	5
x_2	4	6	8	10
PC_1	3.35	1.12	-1.12	-3.35

optimal
primal

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(-1 \times 240) + 0 - (-1 \times 240) = 0$$

Step 1: Primal and Dual are satisfied

(1,5) is optimal

$$(0 \times 1) + 0 - (0 \times 0) = 0$$

optimal